

Xmas homework #2

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 24\text{ V}$; $I_P = 1\text{ mA}$; $R_S = 47\text{ k}\Omega$; $R_C = 12\text{ k}\Omega$, $R_L = 10\text{ k}\Omega$, $R_F = 1.5\text{ M}\Omega$.

Q_1 : $V_{BE} = 0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

Q_2 : $V_{BE} = -0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

$V_T = 25\text{ mV}$.

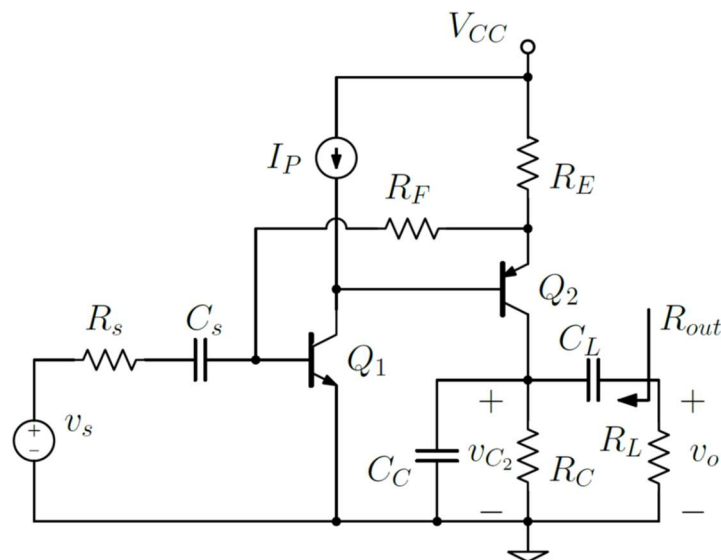


Fig. 1 – A case of two stage amplifier.

Considering all capacitors to be equivalent to open circuits, determine:

1. the value of R_E such that the voltage at the collector of Q_2 is $V_{C2} = 11\text{ V}$ in the steady state;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming C_S and C_L to be equivalent to *short circuits* and C_C to be equivalent to an *open circuit* at the frequencies of interest, determine also:

3. the voltage gain $A_v = v_o/v_s$ of the transistor;
4. the output resistance indicated in the figure.

Finally, considering $C_C = 5\text{ pF}$ not negligible, find

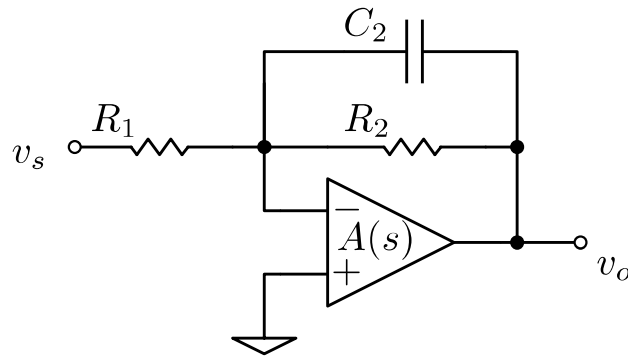
5. the high frequency bandwidth limit of the amplifier (internal capacitances of Q_1 and Q_2 can still be neglected).

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 10 \text{ k}\Omega$; $R_2 = 100 \text{ k}\Omega$; $C_2 = 4.7 \text{ pF}$;

$$A(s) = \frac{10^5}{\left(1 + \frac{s}{10^3}\right)\left(1 + \frac{s}{10^4}\right)} \text{ [V/V]}.$$



Determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. an estimation of the circuit *phase margin*;
3. the value of capacitor C_2 for which the amplifier has 45° phase margin;
4. the complete block diagram of the amplifier, according to *feedback theory*;
5. the expression of the input scaling function $k_{in}(s)$ in the above diagram;
6. the voltage gain at $\omega = 0$ with the specified $A(s)$.